

AI-Based Transformer Health Monitoring and Failure Prediction Using Machine Learning

Presented by:
Madugula Nandini

Department of Electrical and Computer Engineering

Introduction

- Power transformers are essential components of the electrical power distribution system. Transformer failures can cause power outages, equipment damage, and high maintenance costs. Traditional maintenance methods rely on periodic manual inspections, which may not detect faults early. Artificial Intelligence (AI) and Machine Learning (ML) enable predictive maintenance by identifying potential failures in advance. This project uses transformer parameters such as: Load (kW)
- Temperature (°C)
- Voltage (V)
- Oil Level (%)
- Transformer Age (Years)

- A Machine Learning model analyzes these parameters to predict whether a transformer is Healthy or At Risk of Failure.
- A Streamlit web dashboard displays transformer details, prediction results, failure probability, and district-wise monitoring.
- The proposed system helps reduce maintenance costs, improve reliability, and minimize unexpected transformer failures.
- This project demonstrates how AI can support the development of smart and reliable power infrastructure.

Problem Statement

- Manual transformer monitoring is time-consuming and labor-intensive.
- Transformer faults are often detected only after failure occurs.
- Unexpected transformer failures lead to power outages and service interruptions.
- High maintenance and repair costs increase operational expenses.
- There is no intelligent system to predict transformer failures in advance.
- Utilities need a reliable solution for early fault detection and predictive maintenance.
- AI and Machine Learning can analyze transformer data and identify potential failures before they happen.

Objectives

- To monitor the health condition of power transformers.
- To predict transformer failures using Machine Learning algorithms.
- To analyze transformer parameters such as Load, Temperature, Voltage, Oil Level, and Age.
- To develop an interactive web dashboard for transformer monitoring.
- To provide early alerts for transformers at high risk of failure.
- To reduce maintenance costs and improve power system reliability.

Literature Survey

Paper	Author(s)	Method Used	Key Findings	Limitation
Transformer Fault Diagnosis Using Machine Learning	IEEE Research	Machine Learning (Random Forest, SVM)	Predicted transformer faults using operational parameters with good accuracy.	Focused only on fault prediction; no monitoring dashboard.
IoT-Based Smart Transformer Health Monitoring	IEEE Research	IoT Sensors + Cloud Monitoring	Enabled real-time monitoring of transformer parameters such as temperature and voltage.	Required expensive IoT hardware and continuous internet connectivity.
Predictive Maintenance of Power Transformers Using AI	IEEE Research	Artificial Intelligence & Predictive Analytics	Reduced maintenance cost by detecting faults before failure occurred.	Did not provide an interactive visualization dashboard.
Smart Grid Transformer Monitoring System		Combined sensor networks with cloud computing to monitor transformer performance.	Improved reliability of the power distribution system through continuous monitoring.	Focused mainly on data collection and monitoring without predictive analytics.

Methodology

Step 1: Dataset Generation

- Created a **realistic transformer dataset** containing **3001 records**.
- Included important transformer parameters:
 - Transformer ID
 - District
 - Load (kW)
 - Temperature (°C)
 - Voltage (V)
 - Oil Level (%)
 - Age (Years)
 - Failure Status (0 = Healthy, 1 = Fault)

Step 2: Data Preprocessing

- Loaded the dataset into Python using Pandas.
- Checked for missing and duplicate values.
- Selected the required features for model training.
- Split the dataset into **80% training data (2400 records)** and **20% testing data (601 records)**.

Step 3: Machine Learning Model Training

- Trained three Machine Learning models:
 - Logistic Regression
 - Random Forest
 - XGBoost
- Compared their prediction performance.

Step 4: Model Evaluation

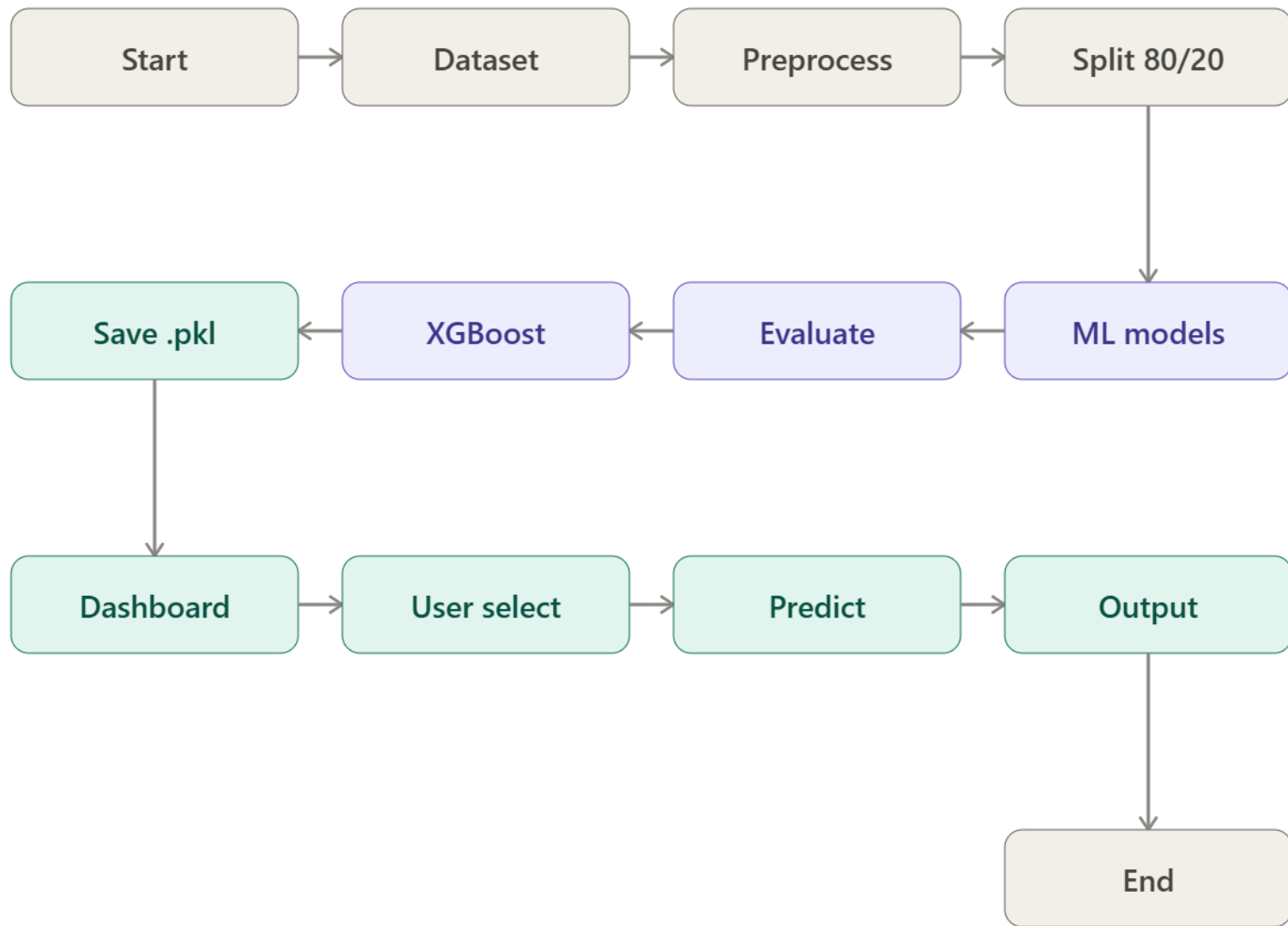
- Evaluated the models using accuracy.
- Selected the best-performing model.
- Saved the trained model as **model.pkl** for future predictions.

Step 5: Dashboard Development

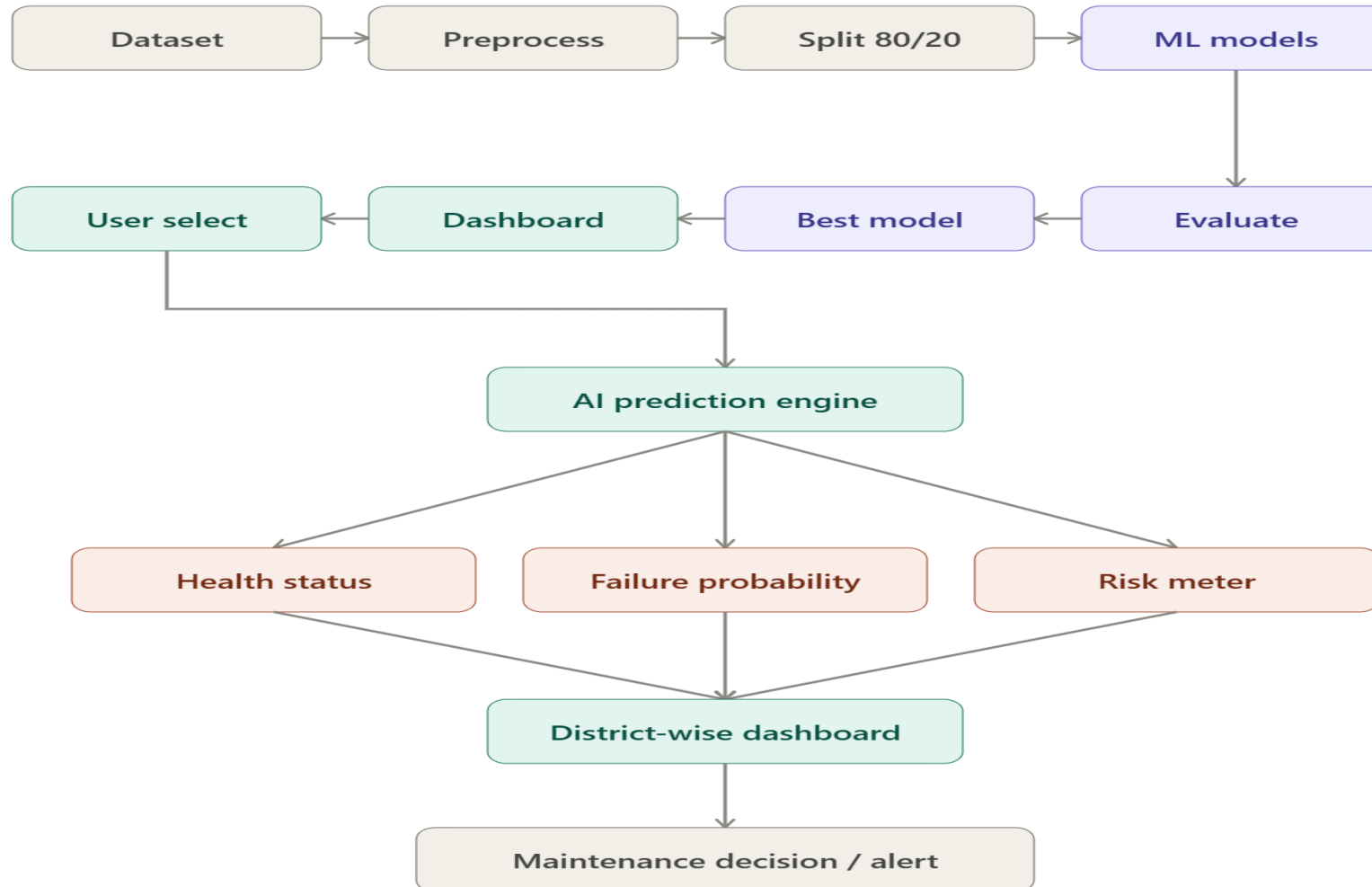
- Developed an interactive dashboard using **Streamlit**.
- Displayed transformer details, prediction results, risk meter, and district-wise summary.

Step 6: Transformer Health Prediction

- The user selects a district and transformer ID.
- The trained model analyzes **Load, Temperature, Voltage, Oil Level, and Age**.
- The system predicts whether the transformer is **Healthy** or **Faulty**.



System Architecture



Dataset Overview

- **Dataset Type:** Simulated Transformer Health Monitoring Dataset
- **Total Records: 3001**
- **Purpose:** To train and test Machine Learning models for predicting transformer health and failure.
- **Districts Covered: 5**
 - Coimbatore
 - Madurai
 - Salem
 - Tiruchirappalli
 - Tenkasi
- **Training Dataset:** 2400 records (80%)
- **Testing Dataset:** 601 records (20%)

- **Target Variable**
- **Failure = 0** → Healthy Transformer
- **Failure = 1** → Faulty / High-Risk Transformer

Machine Learning Models

- Three Machine Learning models were used for transformer failure prediction:
 - **Logistic Regression**
 - **Random Forest**
 - **XGBoost**
- Logistic Regression was used as a baseline classification model.
- Random Forest improved prediction accuracy by combining multiple decision trees.
- XGBoost provided high-performance and efficient fault prediction.
- All models were trained using the transformer dataset.
- The models were evaluated and compared based on their prediction accuracy.
- The best-performing model was selected and saved as **model.pkl** for deployment in the Streamlit dashboard.

Dash board

- https://drive.google.com/file/d/1kY_2CJQqZCEZ482R1OQLbX8uqG8oDgm1/view?usp=drive_link

Control Panel

Select District

tenkasi

Select Transformer ID

18

Predict Transformer Health

IN AI-Based Transformer Health Monitoring & Failure Prediction System



Selected Transformer Details

	Transformer_ID	District	Load_kW	Temp_C	Voltage	Oil_Level	Age_Years	Failure
17	18	tenkasi	87.6756	52.168	237.2508	81.9026	5	0

Select Transformer ID

18

▼

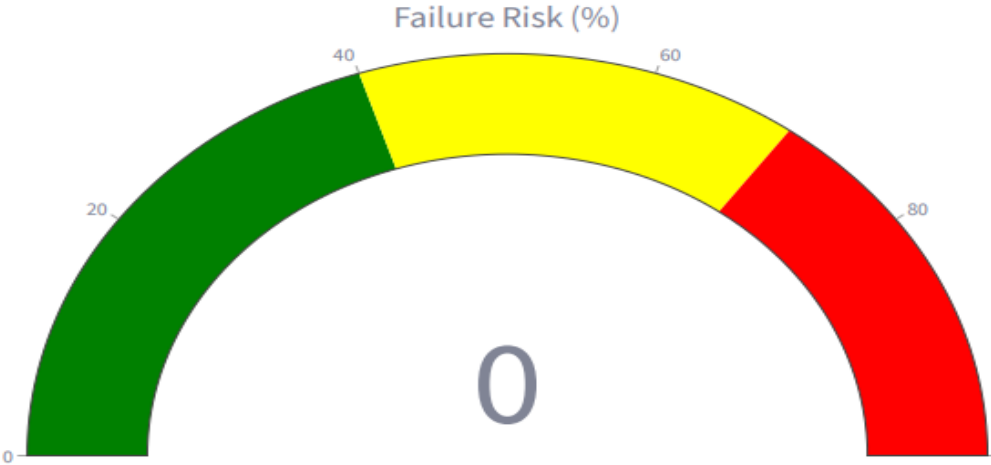
Predict Transformer Health

 Selected Transformer Details

	Transformer_ID	District	Load_kW	Temp_C	Voltage	Oil_Level	Age_Years	Failure
17	18	tenkasi	87.6756	52.168	237.2508	81.9026	5	0

 Prediction Result

 Risk Meter

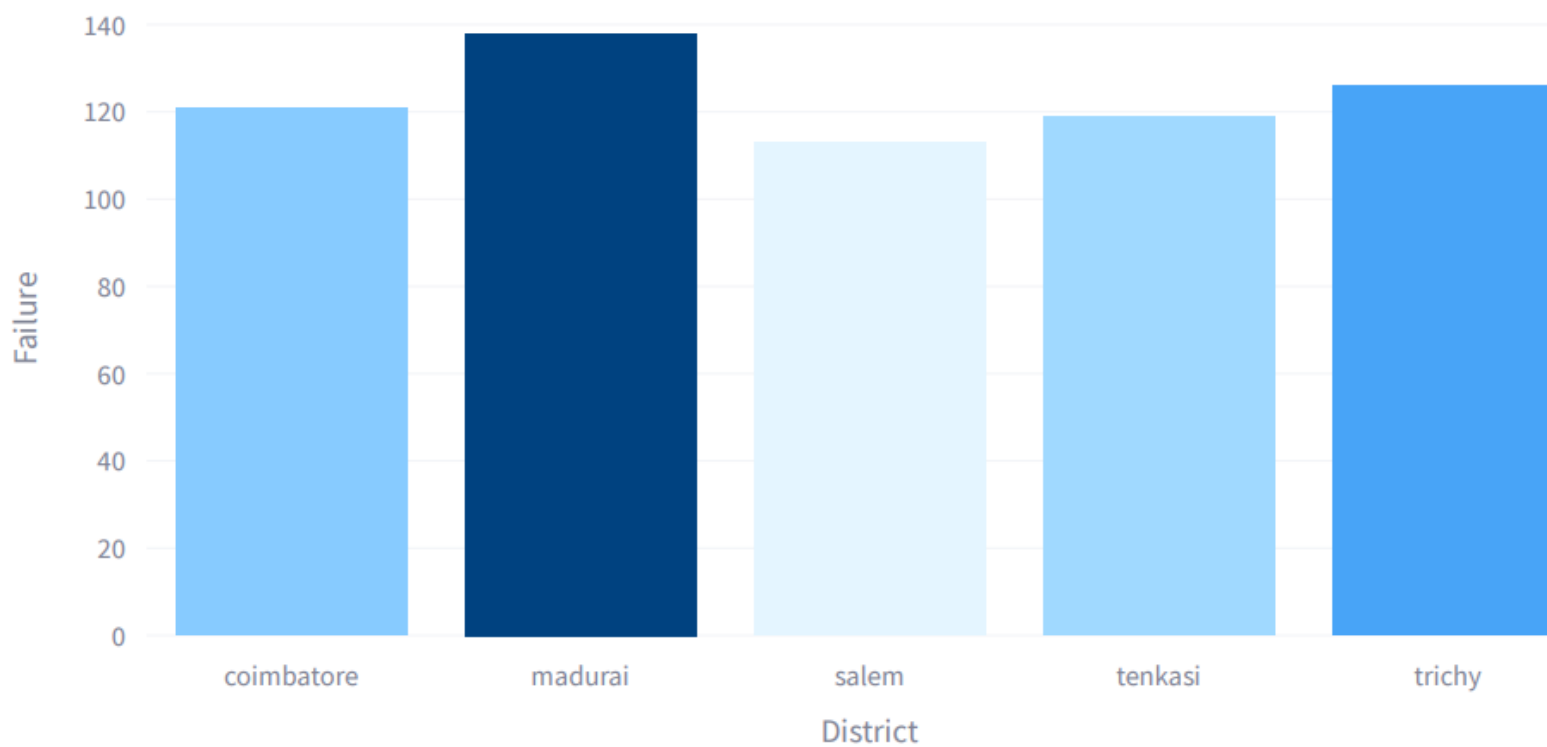


 Transformer Healthy. Failure Probability: 0.00%



District Risk Summary

Number of High-Risk Transformers per District

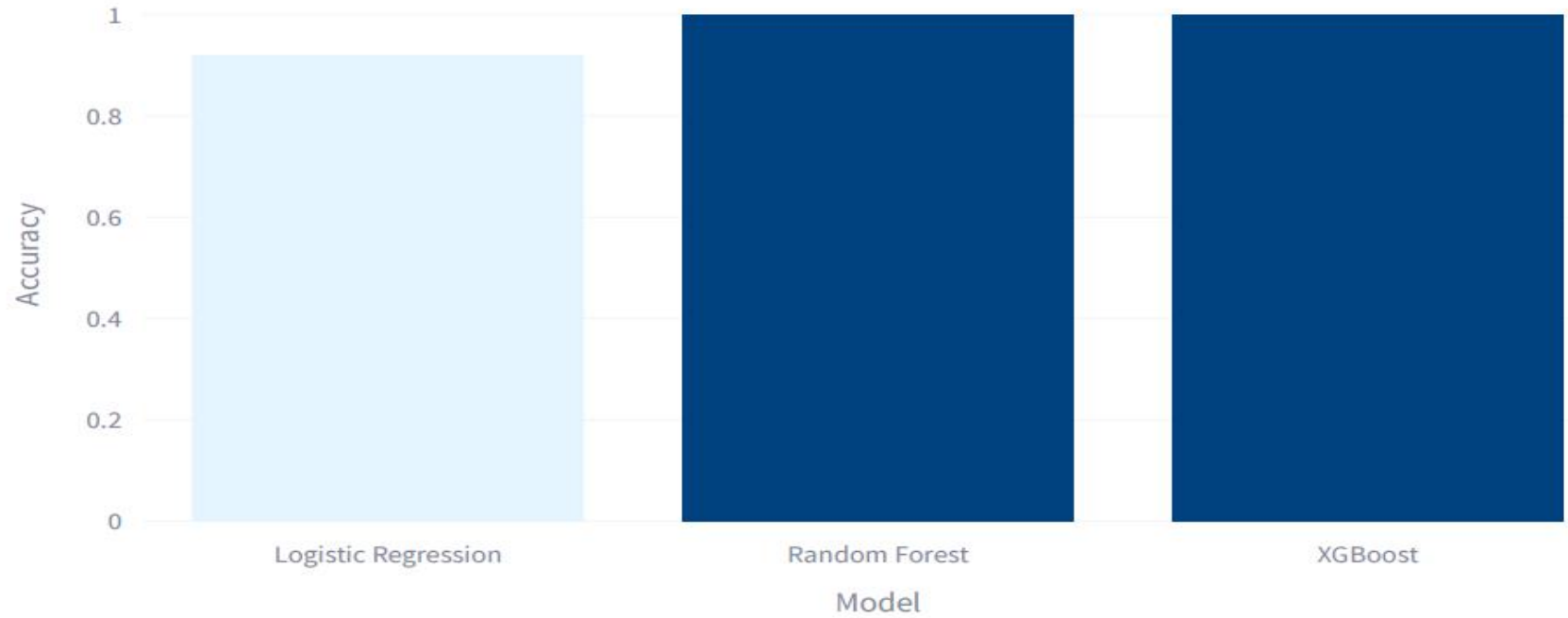


Transformer Monitoring System



Model Accuracy Comparison

Model Accuracy Comparison



Results

- Successfully predicted transformer health using Machine Learning.
- Displayed **Healthy** or **Faulty** status.
- Calculated transformer **Failure Probability**.
- Visualized risk using a **Risk Meter**.
- Displayed district-wise transformer statistics.
- Enabled early identification of high-risk transformers.

Applications

- Power Distribution Companies
- Smart Grid Systems
- Electrical Substations
- Industrial Power Plants
- Renewable Energy Systems
- Utility Monitoring Centers
- Research and Academic Projects

Conclusion

- Developed an AI-based Transformer Health Monitoring and Failure Prediction System.
- Used Machine Learning to predict transformer failures.
- Compared multiple ML models and selected the best-performing model.
- Built a Streamlit dashboard for visualization and monitoring.
- Enabled district-wise transformer health monitoring.
- The proposed system supports predictive maintenance and improves power system reliability.

Thank You